



A physics-guided neural network-based surface potential calculation for GaN HEMT devices

Yunying Zhou, Jun Liu^{*}, Jie Wang

Innovation Center for Electronic Design Automation Technology, Hangzhou Dianzi University, Hangzhou, 310018, China

ARTICLE INFO

Keywords:

Surface potential
Neural network
Physics-guided neural network(PGNN)
GaN HEMT

ABSTRACT

A physics-guided neural network (PGNN) for surface potential calculation in Gallium Nitride (GaN) high-electron-mobility transistors (HEMTs) is proposed in this paper. Since the discontinuities of piecewise analytical function, lack physical consistency in data-driven forms, or rely on empirical Drain-Induced Barrier Lowering (DIBL) parameters to indirectly capture drain-bias effects, we employ a PGNN that directly takes drain voltage (V_d) as input and replaces threshold voltage (V_{th}) with the physically derived $V_{th,dibl}$. Compared with purely data-driven neural networks and piecewise analytical solutions, the PGNN demonstrates superior physical consistency and numerical smoothness, closely matching reference numerical-solution accuracy.

1. Introduction

With the advancement of semiconductor technology, compound-semiconductor-based high-electron-mobility transistors (HEMTs), such as Gallium Nitride (GaN) [1,2] for high-power, high-frequency applications and Gallium Arsenide (GaAs) [3] for high-frequency, low-noise circuits, have become indispensable in advanced electronic systems. Due to strong quantum confinement of channel charge and the non-analytic relationship between the epitaxial structure and charge distribution, accurate HEMT modeling relies on self-consistent Poisson-Schrödinger (SPS) solvers to capture electrostatics and enable physics-based transport models [4]. Advanced surface potential-based compact models such as HiSIM [5,6] and ASM [7] incorporate this underlying physics by embedding quantum-mechanically informed charge-potential relationships into analytically tractable formulations, enabling efficient simulation through the solution of transcendental surface-potential equations.

Traditional approaches to the surface-potential equation rely on iterative numerical schemes, such as the Newton-Raphson method is adopted in HiSIM [5,6] and PSP [8], while the Bisection method is mentioned in Ref. [9]. The accurate results require a reasonable initial guess in order to converge. To avoid iteration, industry-standard models like ASM [7], BSIM6 [10] and MM10 [11] adopt piecewise regional approximations. Complex smoothing methods such as smoothing functions [8,11] and Householder transformation [12] mathematical are applied to piecewise regional functions. Recently proposed unified

regional modeling (URM) frameworks eliminate partitioning but retain equations with multiple exponential terms [13–15]. Beyond mathematical continuity, a robust surface-potential model must also provide a complete physical description. Reference [16] only considers the threshold voltage (V_{th}) under gate voltage (V_g). With decreasing device channel length, high voltage (V_d) weakens gate control, leading to V_{th} roll-off, thereby altering the fundamental gate voltage (V_g) to surface-potential ϕ_s relationship (V_g - ϕ_s) [17,18]. Reference [10] account for V_d indirectly through empirical parameters associated with Drain-Induced Barrier Lowering (DIBL), which require parameter extraction.

In recent years, machine learning has advanced device modeling by offering high efficiency and accuracy [19–21]. For instance, neural networks trained on TCAD data have been used to predict complete I-V/C-V curves and key figures of merit like transconductance and DIBL [19], or to model GAA nanowire FETs with high accuracy (<1.62% error) and significant speedup (>100×) [20]. Reuter et al. leverage data driven neural networks construct Verilog-A compact models, achieving simulation accuracy within 1% error [21]. However, these methods are purely data-driven, learning patterns directly from simulation datasets. Recent work integrates physical knowledge to improve generalizability, such as by employing interpretable Kolmogorov-Arnold Networks (KANs) for FinFET modeling, which offer higher accuracy and symbolic formula derivation from data [22]. The paradigm of Physics-Guided Neural Networks (PGNNs), pioneered by Raissi et al., embeds physical laws as training constraints, thereby improving generalization without

^{*} Corresponding author.

E-mail address: ljun77@hdu.edu.cn (J. Liu).

<https://doi.org/10.1016/j.mejo.2026.107278>

Received 4 March 2026; Received in revised form 18 May 2026; Accepted 18 May 2026

Available online 20 May 2026

1879-2391/© 2026 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

overly complex architectures [23]. Although these physics-guided methods have been applied to device sizing [24], I-V curve fitting [25] and parameter extraction [26], their application to solving the surface potential equation remains relatively limited.

Using the representative surface-potential-based model ASM as an example, this paper proposes a PGNN-based solver that employs weighted loss to compute the surface potential. Unlike purely data-driven models which often lack physical interpretability and exhibit limited extrapolation capability [23], the adoption of PGNN alleviates the inherent discontinuities in piecewise analytical models, thereby preventing spurious derivatives and simulation failures. Besides, unlike the indirect treatment of V_d discussed earlier, the PGNN takes V_d directly as input to enable data-driven learning, while V_{th} is replaced by $V_{th,dibl}$ (Eq. (2)) [27] to embed physical constraints. Although demonstrated on the ASM model with GaN HEMT data, the proposed PGNN framework is applicable to other piecewise compact models for compound semiconductor HEMTs (e.g., GaAs HEMTs), providing a unified design-oriented approach.

2. Theory preparation

Computing the surface potential centers on determining the Fermi level (E_f). Based on the self-consistent Schrödinger-Poisson solution within the triangular potential well approximation, the relationship between sub-band energies E_0 , E_1 and charge density n_s is established. By incorporating Fermi-Dirac statistics and the charge-conservation form of the Poisson equation, Eq. (1) is derived through piecewise approximation [12]. This expression captures the fundamental dependence of E_f on V_g . Although a smoothing function $H(V_{gop})$ is employed to enforce global continuity of the piecewise-defined E_f , it increases computational cost and may introduce spurious derivatives if not properly formulated.

With decreasing device channel length, V_{th} decreases as V_d increases (see Fig. 4 of Reference [17]). In Khandelwal's formulation [12], V_d influences E_f indirectly through an empirical DIBL module, which requires parameter extraction to capture the V_d -induced V_{th} shift. The PGNN approach effectively mitigates the above limitations. To accurately describe this physical mechanism, V_{th} in the PGNN loss equation is replaced with $V_{th,dibl}$ (Eq. (2)) [27]. The ratio B/A determines the DIBL sensitivity, based on the surface-potential distribution model two main approaches proposed by Mura et al. [28] and the 2D analysis by Toyabe et al. [29]. According to Arora's monograph [30] and the classic work by Arora and Richardson [31], the DIBL coefficient can be expressed as shown in Eq. (3).

$$E_{f,unified} = V_{go} \frac{2V_{th} \ln \left(1 + \exp \left(\frac{V_{go}}{2V_{th}} \right) \right)}{\frac{1}{H(V_{gop})} + \frac{C_g}{q \cdot D} \exp \left(\frac{V_{go}}{2V_{th}} \right)} \quad (1)$$

$$V_{th,dibl} = V_{th} \frac{B \cdot V_{ds}}{A} \quad (2)$$

$$A = \frac{1}{\sigma_0 + \sigma_1 \cdot V_{sb}} B = \frac{\epsilon_0 \cdot \epsilon_i}{\pi \cdot C_{ox} \cdot L_g^m} \quad (3)$$

Here, D is the 2D density of states in HEMT, the $V_{go} = V_g - V_{th,dibl}$ is the effective gate voltage, where $V_{th,dibl}$ is the threshold voltage that accounts for the V_d effect. ϵ_0 is the vacuum permittivity, ϵ_i is the relative permittivity of AlGaIn, C_{ox} is the gate oxide capacitance per unit area, L_g is the channel length, m is a constant, typically ranging from 1 to 3, σ_0 are constants, with σ_0 typically set to 0.

In this work, we set $m=1$, $\sigma_0 = 1$, $\sigma_1 = 1$, $L_g=0.25 \mu m$ and body bias $V_{sb} = 0$, yielding $A=1$. The DIBL coefficient then simplifies to:

$$\frac{B}{A} = \frac{\epsilon_0 \cdot \epsilon_i}{\pi \cdot C_{ox} \cdot L_g} \quad (4)$$

This ratio is fully determined by physical constants, material parameters and device geometry, requiring no empirical extraction from measurements or simulations. It should be noted that $V_{th,dibl}$ in Eq. (2) serves as a physics-informed prior in our PGNN framework. Even if the B/A value calculated by Eq. (4) slightly deviates from the true device parameter, the resulting error can be absorbed by the network during training. This combination of physical prior and data-driven learning is the key advantage of our approach over both empirical DIBL models and purely data-driven neural networks.

3. The PGNNs-Based method

This section details the methodology for generating training samples, designing the network architecture, constructing the loss function, and configuring training parameters for the PGNN. Notably, unlike existing PGNN methods that are mostly used for I-V fitting or parameter extraction, this work targets surface-potential calculation for HEMT devices. Accordingly, we design a physics-guided loss function that incorporates quantum effects, statistical physics, and other physical constraints. The DIBL effect is explicitly treated by replacing the V_{th} with a physically derived $V_{th,dibl}$.

3.1. Data generation

The Newton-Raphson method provides high-accuracy solutions suitable for generating neural network training datasets, as in prior studies that employed TCAD-generated data [19–21,32,33]. Fig. 1 illustrates the complete algorithm flow.

Module (a)-Initialization: The process takes gate voltage V_g and drain voltage V_d as inputs, with the Fermi level E_f initialized as:

$$E_f = V_g - V_{th,dibl} - 0.1 \quad (5)$$

Where initialization strategy is grounded in device physics: $V_g - V_{th,dibl}$ represents the gate overdrive, while the $-0.1V$ offset ensures the iteration starts from the sub-threshold region, avoiding numerical oscillations during convergence. In addition, the following control parameters are set once before entering the main iteration loop: the maximum number of iterations is set to $i_{max}=1000$ to prevent infinite loops, the convergence tolerance is set to 10^{-12} for later convergence checking, and the iteration counter is initialized to $i = 0$.

Module (b)-Physical Quantity Calculation: Within the iteration loop, the current estimate of E_f (initialized in Module (a) and updated iteratively in Module (d)) is used to compute electron density $n_{s,elec}$ through the charge-conserving Poisson equation. If $n_{s,elec} > 0$, the quantum-confined subband energies E_0 and E_1 are calculated using the triangular potential well approximation (Eq. (6)). Otherwise, they are set to zero.

$$E_i = \gamma_i \cdot n_{s,elec}^{2/3} \quad (i = 0, 1, \gamma_0 \text{ and } \gamma_1 \text{ extracted from [12]}) \quad (6)$$

Module (c)-Convergence Check: Using $n_{s,elec}$ obtained from Module (b) and $n_{s,stat}$ evaluated from Fermi-Dirac statistics, and the loss function $F(E_f) = n_{s,elec} - n_{s,stat}$ is constructed. Convergence is reached if $|F(E_f)| < 10^{-12}$, the numerical solution is output.

Module (d)-Iterative Update: If convergence is not achieved, the derivative dF/dE_f is computed via the forward difference method, and E_f is updated using the Newton update rule:

$$E_f^{(k+1)} = E_f^{(k)} - F(E_f^{(k)}) / (dF/dE_f) \quad (7)$$

The iteration is limited to 1000 steps. The updated E_f is then fed back to Module (b) for the next iteration. While this physically rigorous numerical approach yields high-precision reference solutions for surface potential, its convergence is sensitive to the initial guess, which can lead to divergence.

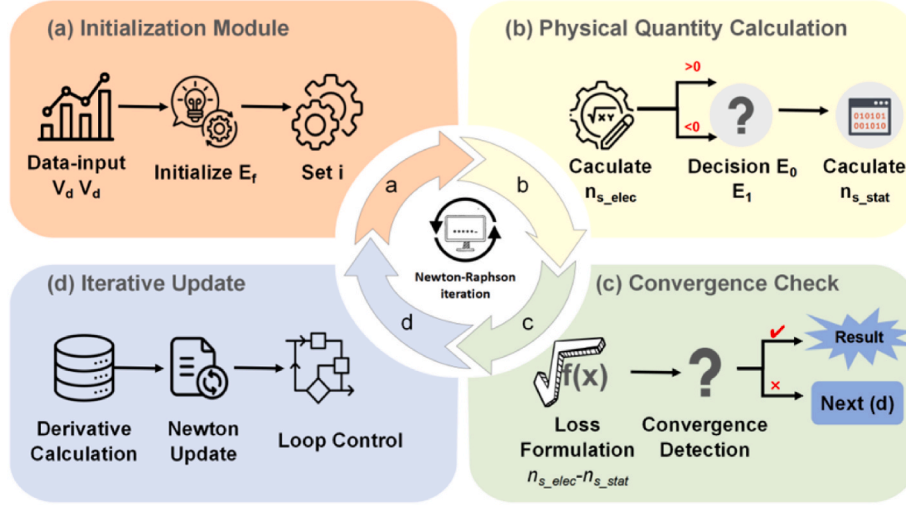


Fig. 1. The Newton iteration calculation process. The solver takes gate voltage V_g and drain voltage V_d as inputs. Module (a) is executed once for initialization, while Modules (b)-(d) form the iterative loop that repeats until convergence or the maximum iteration count (1000) is reached.

3.2. PGNN neural network

This paper constructs a PGNN using a fully connected feed forward architecture, as shown in Fig. 2, the input gate voltage V_g and drain voltage V_d , to four physical outputs, i.e. $f: V_g, V_d \rightarrow (E_0, E_1, E_f, \log(n_s))$. The network accepts V_g ranging from -1V to 4V and V_d from 0V to 1V as inputs. The model outputs four physical quantities: $(E_0, E_1, E_f, \log(n_s))$. The architecture includes three hidden layers, each with 128 neurons. Since the carrier density n_s spans a wide dynamic range up to 10^{17} , the network outputs $\log(n_s)$ to maintain numerical stability during training. The actual physical value is later restored using $n_s = \exp(\log(n_s))$. The hidden layers use the LeakyReLU activation function with a negative slope of 0.01. This choice is based on the continuous nonlinear variation of physical parameters in HEMT devices when the gate voltage sweeps from the below-threshold ($V_g < V_{th,dibl}$, off-state) to the above threshold ($V_g > V_{th,dibl}$, on-state). LeakyReLU preserves a small gradient in the negative region, allowing the network to accurately capture physical behavior across the full operating range while avoiding the gradient vanishing problem that can occur with standard ReLU in complex physics-constrained network.

3.3. The loss function

The PGNN embeds physical constraints into its training via four physics based loss terms (L_{ns} , L_{Ef} , L_{E0} , L_{E1}), a supervised loss term L_{sup} is incorporated to accelerate convergence. L_{ns} is the loss derived from Fermi-Dirac statistics, where $\hat{n}_{sl}^{(i)} = n_{s,feim}(\hat{E}_0^{(i)}, \hat{E}_1^{(i)}, \hat{E}_f^{(i)})$ represents the carrier density computed from Fermi-Dirac statistics (Eq. (8)). This loss enforces that the predicted energy band parameters conform to Fermi-Dirac statistics. L_{Ef} is the charge conservation loss based on the charge-balanced Poisson equation, where $\hat{E}_f^{(i)} = E_{f,charge}(V_g^{(i)}, \hat{n}_s^{(i)}, V_d^{(i)})$ denotes the Fermi level inversely computed from (Eq. (9)). This term ensures that the predicted carrier density $\hat{n}_s$ and Fermi level $\hat{E}_f$ vary with the gate voltage V_g . L_{E0} , L_{E1} are the sub band energy losses derived from the Schrödinger equation, where $\hat{E}_{01}^{(i)} = E_{0,ns}(\hat{n}_s^{(i)})$, $\hat{E}_{11}^{(i)} = E_{1,ns}(\hat{n}_s^{(i)})$ are the energy levels computed using the predicted carrier density (Eq. (10)). The supervised loss L_{sup} measures direct prediction error without physical constraints. The total loss function is defined as the weighted sum of all individual loss terms. The mathematical expressions for each loss function and the total loss are given in Eqs. 11, 12, and 13.

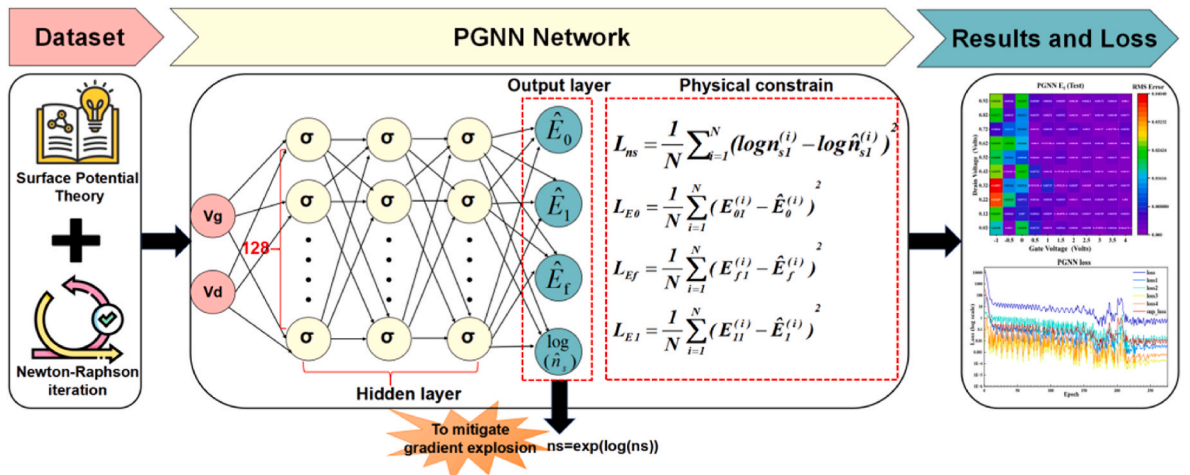


Fig. 2. The physics-guided neural network.

$$n_s = D \cdot \frac{kT}{q} \left[\ln \left(1 + e^{\left(\frac{q}{kT} (E_f - E_1) \right)} \right) + \ln \left(1 + e^{\left(\frac{q}{kT} (E_f - E_0) \right)} \right) \right] \quad (8)$$

$$n_s = \frac{\varepsilon}{q \cdot d} (V_{g0} - E_f) = \frac{C_g}{q} (V_{g0} - E_f) \quad (9)$$

$$E_i = \gamma_i \cdot n_s^{2/3} \quad (i = 0, 1) \quad (10)$$

$$L_{sup} = \sum MSE(\hat{y}_k, y_k), y_k \in \{E_0, E_1, E_f, \log n_s\} \quad (11)$$

$$L_k = \frac{1}{N} \sum_{i=1}^N (y_k^{(i)} - \hat{y}_k^{(i)})^2, y_k \in \{n_s, E_f, E_0, E_1\} \quad (12)$$

$$L_{total} = \sum_{i=1}^4 w_i \cdot L_k + w_{sup} \cdot L_{sup} \quad (13)$$

Since the physical equations for the four loss functions Eqs.11, 12, and 13 are physically coupled, a significant increase in any individual loss value indicates a deviation in the corresponding predicted variable ($\hat{E}_0$, $\hat{E}_1$, $\hat{E}_f$ or $\hat{n}_s$). The four loss functions are harmonized through a normalization mechanism to achieve dimensional consistency, forming a closed-loop optimization system.

3.4. Model training

The Newton iteration-generated samples size is $V_g \cdot V_d = 5001 \cdot 101 = 505,101$ samples. The dataset is randomly split into three subsets: 70% for training, 15% for validation and 15% for testing. The training set is used to update the network weights. The validation set is used for monitoring training progress, early stopping, and learning rate scheduling. The test set is used only for final evaluation and does not participate in any hyperparameter tuning or model selection.

The PGNN model was optimized using the Adam optimizer with an initial learning rate of 0.001, coupled with a Multi Step learning rate scheduler. The batch size was set to 406*8, and the maximum number of training epochs was 1000. Early stopping was applied with a patience of 100.

4. Testing result

Using the Newton-Raphson numerical solution as a benchmark, this paper evaluates the device modeling performance of the PGNN. The results show that the PGNN offers superior predictive capability, especially near the $V_g = V_{th,dibl}$ with significantly lower error compared to purely data-driven approaches.

4.1. Comparison with conventional neural network(NN)

The termination of PGNN training required 98 and 343 s for the V_g -input and $V_g V_d$ -input models, respectively, with the dataset size increased by 1000 times. No existing neural network has been developed for surface potential calculation in HEMT devices. We therefore use a conventional neural network as a baseline. This network shares the same architecture and training data as the PGNN, but its loss function includes only the supervised loss L_{sup} without any physics constraints(L_{ns} , L_{Ef} , L_{E0} , L_{E1}), allowing us to evaluate the performance gain brought by physics guidance.

Table 1 shows comparable performance between the PGNN and the purely data-driven Conventional NN with only V_g as input. However, when both V_g and V_d are applied, the introduction of V_d allows the surface-potential equation to better capture the two-dimensional potential distribution, markedly increasing its nonlinearity and variable

Table 1

MSE comparison of output parameters on the **training dataset**. The PGNN exhibits up to a 100-fold reduction in MSE compared to conventional neural networks.

(a) V_g -input.(Conventional NN and PGNN converged after 480 and 780 training epochs, respectively.)				
Parameter	Conventional NN	PGNN	Conventional NN ($V_g = V_{th}$)	PGNN ($V_g = V_{th}$)
E_0	2.631e-05	1.890e-07	4.753e-06	1.105e-06
E_1	3.836e-05	7.821e-06	1.676e-06	7.061e-06
E_f	6.578e-05	2.426e-05	4.348e-05	5.445e-05
$\log(n_s)$	0.084	0.042	0.015	0.011
(b) $V_g V_d$ -input.(Conventional NN and PGNN converged after 135 and 548 training epochs, respectively.)				
Parameter	Conventional NN	PGNN	Conventional NN ($V_d = 0.02$ $V_g = V_{th,dibl}$)	PGNN ($V_d = 0.02$ $V_g = V_{th,dibl}$)
E_0	3.355e-06	1.406e-07	6.071e-06	5.440e-08
E_1	1.332e-05	3.535e-07	8.310e-06	1.412e-07
E_f	1.360e-05	2.179e-06	5.332e-06	2.386e-07
$\log(n_s)$	0.0174	0.0019	0.0555	0.0024

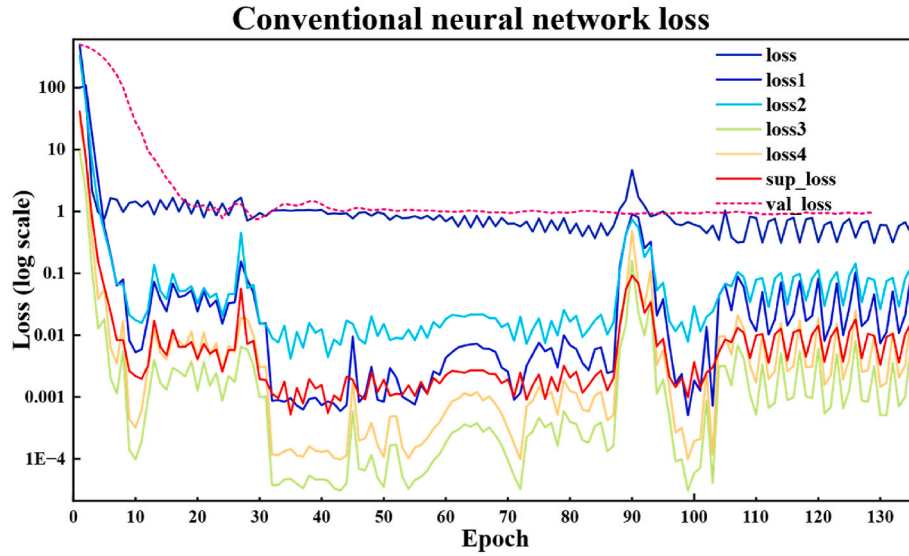
coupling. Under these conditions, the PGNN achieves significantly higher predictive accuracy than the conventional neural network. Fig. 3 shows the training and validation dataset loss trends of the two neural network types versus the number of iterations. Fig. 4 presents a comparative analysis of the output characteristics for both neural networks using test set data. It can be observed that the conventional neural network, as a purely data-driven approach without physical constraints, converges faster, but the PGNN achieves higher accuracy. The accuracy advantage of the PGNN demonstrated is not merely numerical. Surface potential is the foundation of compact device models. As established in Reference [34], once the surface potential is accurately determined, the channel charge density, terminal currents, and capacitances follow directly. The threshold voltage V_{th} , extracted from the E_f - V_g characteristic, is a critical parameter for HEMTs. It determines the on/off switching behavior of the transistor [35], as well as the noise margin in digital circuits [36] and the bias point in analog circuits [37].

4.2. Comparison with analytical model

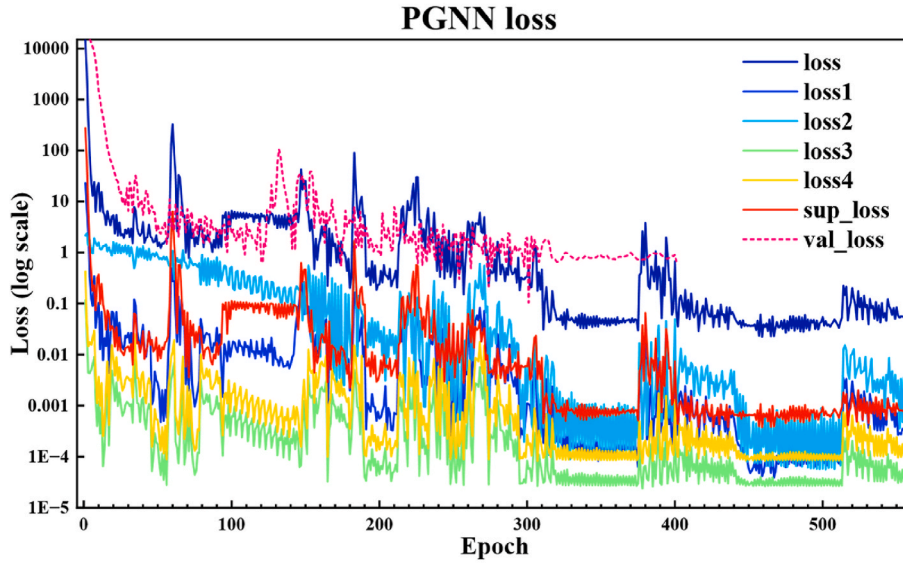
To enable a fair and reliable comparison, this study implements the piecewise analytical model of Khandelwal [12]. In the critical region where the gate voltage approaches the threshold voltage ($V_g \approx V_{th,dibl}$, $V_d = 0.01V$), numerical methods such as the Householder algorithm are introduced. However, this high accuracy comes at the expense of significantly increased model complexity and computational cost. As shown in Fig. 5, near $V_g \approx V_{th,dibl}$ the PGNN yields a smooth E_f profile with an RMS error 40× smaller than the analytical solution, without any empirical DIBL correction or smoothing. Furthermore, SPICE simulators demand continuous and differentiable model behavior. The PGNN delivers a smooth surface potential near threshold, which naturally avoids abrupt changes in C-V curves. This is a key condition for stable simulation convergence and accurate prediction of transconductance, gain, and efficiency.

4.3. Ablation study and sensitivity analysis

To quantify the contribution of each physical constraint, we start with a purely data-driven neural network(using only L_{sup}) and then add the physics-guided loss terms one by one. Table 2 shows that adding



(a) The Conventional neural network loss.



(b) The PGNN loss.

Fig. 3. The training and validation dataset loss trends of the two neural networks type. (a) The Conventional neural network loss (converged at epochs 135/129 for training/validation loss, respectively). (b) The PGNN loss (converged at epochs 548/401 for training/validation loss, respectively).

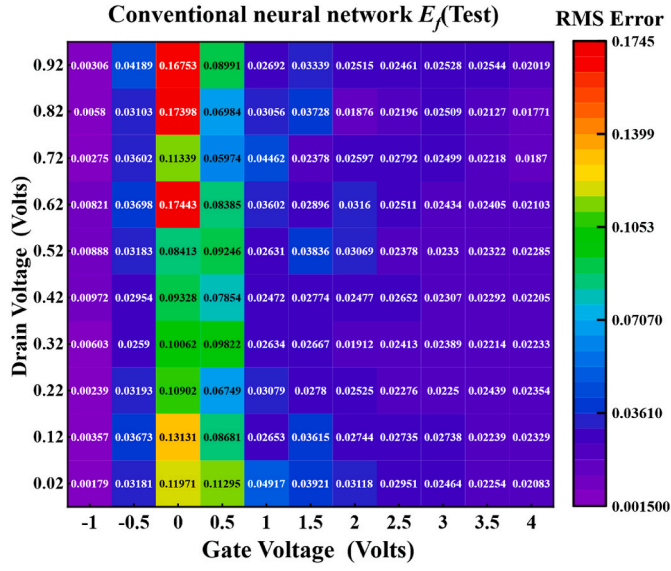
each physical constraint consistently reduces the E_f prediction error. L_{ns} directly couples the Fermi level E_f to the carrier density n_s , which lies at the core of the surface potential equation. Consequently, when L_{ns} derived from Fermi-Dirac statistics is included, the E_f error drops significantly (by a factor of 6). The two subband energy terms, L_{E0} and L_{E1} , bring further but smaller improvements, consistent with their role as secondary quantum constraints.

Using the loss weights of Table 2 model #5 as the baseline, we vary the weight of L_{ns} to 0.5 times and 2 times of its original value. Selecting w_1 (associated with the core term L_{ns}) and w_3 (associated with the secondary constraint L_{E0}) for sensitivity analysis. L_{ns} indirectly affects all energy levels via Fermi-Dirac statistics, with the ground state E_0 dominating the channel charge contribution. As shown in Table 3, Reducing w_1 by half leads to a clear increase in E_f error, indicating that L_{ns} requires sufficient weight to effectively constrain the network, doubling it has a much smaller effect, suggesting that the baseline weight is already near optimal. In contrast, varying w_3 has a much weaker impact, consistent

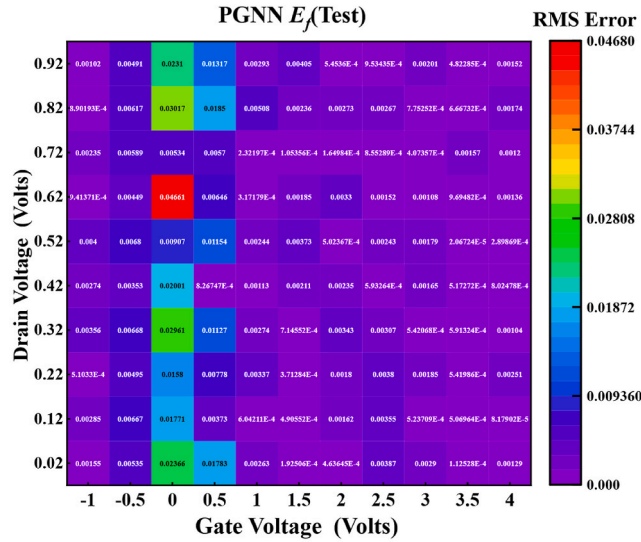
with its secondary role.

4.4. Robustness to input noise

To evaluate the PGNN under non-ideal conditions, we add Gaussian noise with standard deviations of 1%, 2%, 3%, and 5% to V_g and V_d , and perform inference using the pretrained model without retraining. As shown in Table 4, at 1% noise, both E_f MSE and logns MSE increase only slightly, indicating that the PGNN is insensitive to mild input noise. At 2% noise and above, the error grows more noticeably. However, even under 5% noise, the E_f MSE remains far below that of the conventional NN under noise-free conditions. Overall, the PGNN performs reliably up to 1% input noise and maintains acceptable accuracy under higher noise levels.



(a) The conventional NN test error.



(b) The PGNN test error.

Fig. 4. Heat map of E_f test dataset results for two neural networks. The PGNN reduces RMS error by up to two orders of magnitude versus conventional NN in V_g , V_d -input mode.

5. Conclusion

This work presents a PGNN method for surface-potential calculation. It achieves excellent agreement with numerical solutions obtained via Newton-Raphson iteration of the surface-potential equation, while providing smoother and more continuous outputs than piecewise analytical models. By directly incorporating V_d as an input and enforcing physical consistency through $V_{th,dibl}$, this strategy not only alleviates the burden of empirical parameter extraction, but also achieves higher prediction accuracy than purely data-driven counterparts.

In conclusion, the PGNN method not only inherently constrains the model to physically plausible outputs, thereby improving generalization across the device's operational range, but also avoids reliance on closed-form derivations or piecewise approximations, enabling straightforward extension to emerging post-Moore devices.

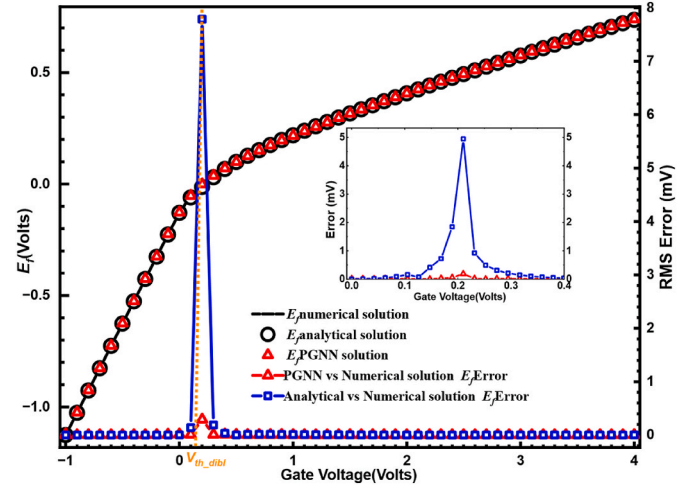


Fig. 5. Absolute error of analytical approximation of surface potential by our method compared to numerical solutions. (Fixed $V_d = 0.01$ V, the PGNN achieves a smooth transition near $V_{th,dibl}$.)

Table 2

MSE for E_f and $\log(n_s)$ at each stage on the validation dataset. (All models are trained for 300 fixed epochs with the loss weights of the full PGNN (#5) and no early stopping.).

Model	L_{sup}	L_{ns}	L_{E_f}	L_{E_0}	L_{E_1}	E_f (MSE)	$\log(n_s)$ (MSE)
#1	✓					1.428e-04	0.1952
#2	✓	✓				2.273e-05	0.1586
#3	✓	✓	✓			8.124e-06	0.0235
#4	✓	✓	✓	✓		5.996e-06	0.0061
#5	✓	✓	✓	✓	✓	2.708e-06	0.0023

Table 3

MSE comparison of output parameters on the validation dataset (trained for 300 epochs).

Varied weight	Relative to baseline	E_f (MSE)	$\log(n_s)$ (MSE)
$W_1(L_{n_s})$	0.5 ×	4.782e-06	0.0026
	1.0 × (baseline)	2.708e-06	0.0023
	2.0 ×	2.783e-06	0.0022
$W_3(L_{E_0})$	0.5 ×	3.051e-06	0.0024
	1.0 × (baseline)	2.708e-06	0.0023
	2.0 ×	2.812e-06	0.0023

Table 4

PGNN prediction error under different input noise levels on the test dataset. (The model is trained for 300 epochs with the same configuration as the full PGNN (#5) in Table 2.).

Noise level	E_f (MSE)	$\log(n_s)$ (MSE)
0%	2.415e-06	0.0023
1%	2.483e-06	0.0028
2%	1.125e-05	0.0033
3%	1.143e-05	0.0037
5%	1.667e-05	0.0051

6. Future work

This work considers a device with a fixed $L_g = 0.25$ μm , where L_g serves as a fixed parameter rather than a PGNN input. Our framework can be readily transferred to other L_g values by recalculating B/A , regenerating the training data and retraining the model. In future work, we will incorporate L_g as an additional input variable to build a unified

PGNN model covering multiple channel lengths. A single trained model will then be applicable to devices with different L_g values, retaining the key advantage of avoiding empirical parameter extraction. This will make the proposed framework more general and practical for compact modeling of GaN HEMTs and other compound semiconductor devices.

Beyond architectural extensions, we will implement the trained PGNN in Verilog-A, allowing it to be directly used in SPICE or Cadence. Circuit designers will then be able to call it like a standard compact model without requiring any knowledge of the underlying neural network.

CRedit authorship contribution statement

Yunying Zhou: Data processing, Model training, Validation, Writing-original draft. Jun Liu: Supervision, Writing-review & editing. Jie Wang: Conceptualization, Writing-review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

References

- [1] U.K. Mishra, L. Shen, T.E. Kazior, et al., GaN-Based RF power devices and amplifiers, *Proc. IEEE* 96 (2) (2008) 287–305.
- [2] R. Rupp, T. Laska, O. Häberlen, et al., Application specific trade-offs for WBG SiC, GaN and high end Si power switch technologies, in: 2014 IEEE International Electron Devices Meeting, IEEE, 2014, 2.3. 1–2.3. 4.
- [3] T. Mimura, S. Hiyaizumi, T. Fujii, et al., A new field-effect transistor with selectively doped GaAs/n-AlxGa1-xAs heterojunctions, *Jpn. J. Appl. Phys.* 19 (5) (1980) L225.
- [4] R. Drury, C.M. Snowden, A quasi-two-dimensional HEMT model for microwave CAD applications, *IEEE Trans. Electron. Dev.* 42 (6) (2002) 1026–1032.
- [5] M. Suetake, K. Suematsu, H. Nagakura, et al., HiSIM: a drift-diffusion-based advanced MOSFET model for circuit simulation with easy parameter extraction, in: 2000 International Conference on Simulation Semiconductor Processes and Devices, IEEE, 2000, pp. 261–264.
- [6] Y. Okada, T. Mizoguchi, Y. Tanimoto, et al., HiSIM GaN: compact model for GaN-HEMT with accurate dynamic current-collapse reproduction, *IEEE Trans. Electron. Dev.* 66 (1) (2018) 106–115.
- [7] S. Ghosh, S.A. Ahsan, A. Dasgupta, et al., GaN HEMT modeling for power and RF applications using ASM-HEMT, in: 2016 3rd International Conference on Emerging Electronics (ICEE), IEEE, 2016, pp. 1–4.
- [8] G. Gildenblat, X. Li, W. Wu, et al., PSP: an advanced surface-potential-based MOSFET model for circuit simulation, *IEEE Trans. Electron. Dev.* 53 (9) (2006) 1979–1993.
- [9] T.L. Chen, G. Gildenblat, Analytical approximation for the MOSFET surface potential, *Solid State Electron.* 45 (2) (2001) 335–339.
- [10] Y.S. Chauhan, S. Venugopalan, M.A. Chalkiadaki, et al., BSIM6: Analog and RF compact model for bulk MOSFET, *IEEE Trans. Electron. Dev.* 61 (2) (2013) 234–244.
- [11] R. Van Langevelde, F.M. Klaassen, An explicit surface-potential-based MOSFET model for circuit simulation, *Solid State Electron.* 44 (3) (2000) 409–418.
- [12] S. Khandelwal, Y.S. Chauhan, T.A. Fjeldly, Analytical modeling of surface-potential and intrinsic charges in AlGaIn/GaN HEMT devices, *IEEE Trans. Electron. Dev.* 59 (10) (2012) 2856–2860.
- [13] X. Zhou, G. Zhu, G.H. See, et al., Unification of MOS compact models with the unified regional modeling approach, *J. Comput. Electron.* 10 (1) (2011) 121–135.
- [14] S. Rustagi, L.C. Hsia, S. Chu, et al., Towards unification of MOS compact models with the unified regional approach, in: 2006 8th International Conference on Solid-State and Integrated Circuit Technology Proceedings, IEEE, 2006, pp. 1193–1197.
- [15] W. Jie, S. Lingling, L. Jun, et al., A surface-potential-based model for AlGaIn/AlN/GaN HEMT, *J. Semiconduct.* (9) (2013) 34.
- [16] S. Choi, D.G. Park, M.J. Kim, et al., Automatic prediction of MOSFETs threshold voltage by machine learning algorithms, in: 2023 7th IEEE Electron Devices Technology & Manufacturing Conference (EDTM), IEEE, 2023, pp. 1–3.
- [17] B. Hult, J. Bergsten, R.F.D. Del Castillo, et al., Characterization of drain-induced barrier lowering in GaN HEMTs using a drain Current injection technique, *IEEE Trans. Electron. Dev.* 71 (2024) 7383–7389.
- [18] A. Malik, N. Jain, M. Mishra, et al., Analytical model to evaluate threshold voltage of GaN based HEMT involving nanoscale material parameters, *Superlattice. Microsc.* 152 (2021) 106834.
- [19] K. Mehta, H.Y. Wong, Prediction of FinFET current-voltage and capacitance-voltage curves using machine learning with autoencoder, *IEEE Electron Device Lett.* 42 (2) (2020) 136–139.
- [20] X. Tang, Z. Li, L. Zeng, et al., Device modeling based on cost-sensitive densely connected deep neural networks, *IEEE J. Electron Devices Soc.* 12 (2024) 619–626.
- [21] M. Reuter, J. Wilm, A. Kramer, et al., Machine learning-based compact model design for reconfigurable FETs, *IEEE J. Electron Devices Soc.* 12 (2024) 310–317.
- [22] R. Novkin, H. Amrouch, Kolmogorov–Arnold network for transistor compact modeling, *Adv. Intell. Syst.* (2025) 2500325.
- [23] M. Raissi, P. Perdikaris, G.E. Karniadakis, Physics-informed neural networks: a deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations, *J. Comput. Phys.* 378 (2019) 686–707.
- [24] Q.H. Kim, S. Lee, A. Ma, et al., A simulation physics-guided neural network for predicting semiconductor structure with few experimental data, *Solid State Electron.* 201 (2023) 108568.
- [25] S. Mishra, B. Gaikwad, N. Chaturvedi, Semi-supervised physics guided deep learning framework: An application in modeling of gallium nitride based high electron mobility transistors, *J. Appl. Phys.* 132 (2022) 1–13, 044902.
- [26] Y. Wei, W. Xing, T.J. Lin, et al., Knowledge transfer for GaN HEMTs parameter extraction based on hybrid model, in: 2024 2nd International Symposium of Electronics Design Automation (ISED), IEEE, 2024, pp. 684–689.
- [27] S.G. Chamberlain, S. Ramanan, Drain-induced barrier-lowering analysis in VLSI MOSFET devices using two-dimensional numerical simulations, *IEEE Trans. Electron. Dev.* 33 (11) (1986) 1745–1753.
- [28] Y. Mura, K. Ohwada, Threshold voltage theory for a short-channel MOSFET using a surface-potential distribution model, *Solid State Electron.* 22 (12) (1979) 1045–1051.
- [29] T. Asai Toyabe, et al., Analytical models of threshold voltage and breakdown voltage of short-channel MOSFETs derived from two-dimensional analysis, *Electron Devices IEEE Transactions on* 26 (1979) 453–461.
- [30] N. Arora, MOSFET Models for VLSI Circuit Simulation: Theory and Practice, Springer-Verlag New York, Inc., 1993.
- [31] N.D. Arora, L.M. Richardson, Mosfet modeling for circuit simulation, *VLSI Electron. Microstruct. Sci.* 18 (1989) 237–276.
- [32] Z. Yao, Z. Hu, P. Lai, et al., Etching process prediction based on cascade recurrent neural network, *Eng. Appl. Artif. Intell.* 139 (2025) 109590.
- [33] C. Liu, J. Wang, S. You, et al., Machine-learning-assisted EEHEMT compact modeling of GaN HEMTs, *Microelectron. J.* 165 (2025) 106861.
- [34] Y. Chen, J. He, Y. He, et al., A surface potential based compact model for GaN HEMT I-V and CV simulation, 2020 IEEE International Conference on Advances in Electrical Engineering and Computer Applications (AEECA) (2020) 490–495.
- [35] J. Chen, M. Hua, J. Wei, et al., OFF-state drain-voltage-stress-induced VTH instability in schottky-type p-GaN Gate HEMTs, *IEEE J. Emerg. Sel. Top. Power Electron.* 9 (2020) 3686–3694, PP(99):1–1.
- [36] Y. Jiang, F.Z. Du, K.Y. Wen, et al., Threshold voltage modulation on a CTL-based monolithically integrated E/D-mode GaN inverters platform with improved voltage transfer performance, *Appl. Phys. Lett.* 125 (3) (2024) 7.
- [37] C. Gaquiere, Impact of multi-bias on the performance of 150 nm GaN HEMT for high-frequency applications, *Micromachines* 16 (2025) 932.